

Predicting ATP Tennis Match Outcomes: Comparing Logistic Regression, Elo Ratings, and Hybrid Models Final Report

Himanshu Muley
A1895211

May 7, 2026

Report submitted for **Data Science Research Project Part B** at
the School of Mathematical Sciences, University of Adelaide



THE UNIVERSITY
of ADELAIDE

Project Area: **Sports Prediction Analytics**
Project Supervisor: **Mikaela Fudolig**

In submitting this work I am indicating that I have read the University's Academic Integrity Policy. I declare that all material in this assessment is my own work except where there is clear acknowledgement and reference to the work of others.

I give permission for this work to be reproduced and submitted to other academic

staff for educational purposes.

OPTIONAL: I give permission this work to be reproduced and provided to future students as an exemplar report.

Abstract

This project investigates how rating-based and statistical learning models compare in their ability to predict professional ATP tennis match outcomes. Using 60,154 matches from the ATP tour between 2000 and 2019, we fit twelve models spanning three paradigms: logistic regression using ATP ranking-point differences, Elo rating systems with fixed and surface-specific update factors, and hybrid models that embed Elo rating differences as covariates within a logistic regression framework. All models are evaluated on a temporally held-out 2019 test set ($n = 2,680$) using accuracy, log-loss, Brier score, calibration curves, and ROC analysis. The logistic baseline achieves the best calibration (log-loss = 0.655, Brier = 0.229) while maintaining competitive accuracy (61.4%). Hybrid models match this accuracy but exhibit substantially worse calibration (log-loss ≥ 0.730), and adding surface type, tournament level, or interaction terms provides no statistically significant improvement once a player-strength predictor is included. These findings suggest that, for the purpose of well-calibrated match-outcome prediction, a simple ranking-point logistic model outperforms more complex hybrid specifications on the ATP tour.

1 Introduction

1.1 Background

Predicting the outcome of professional tennis matches is a well-studied problem in sports analytics that sits at the intersection of rating systems, statistical modelling, and probabilistic forecasting. Unlike team sports, tennis is an individual contest in which the relative strength of two players can, in principle, be captured by a single scalar difference. This structural simplicity has made tennis a natural test sport for comparing prediction methodologies.

Two broad approaches dominate the literature. The first uses *statistical learning models*, typically logistic regression - where match outcomes are modelled as a function of observable predictors such as ranking-point differences, surface type, and tournament tier [3]. The second relies on *rating systems*, most notably the Elo system originally developed for chess, in which each player carries a numerical rating that updates dynamically after every match. S.A. Kovalchik [4] conducted a comprehensive comparison of prediction methods on ATP data and found that Elo-family models and regression-based models achieve broadly similar accuracy, typically in the range of 60–70% depending on the feature set.

A natural question is whether these two approaches can be *combined*. Elo ratings capture cumulative match-by-match form that updates dynamically, while ATP ranking points reflect official standings accumulated over a rolling twelve-month window. A player returning from injury, for example, may retain high ranking points from previous results while their Elo rating has declined due to recent losses. Hvattum and Arntzen [2] were among the first to use Elo rating differences as covariates inside a logistic regression model for football, demonstrating that the Elo difference can serve as a meaningful predictor within a regression framework. More recently, Angelini et al.[1] proposed a Weighted Elo system for tennis that accounts for player form, and compared its performance against logistic regression across large ATP and WTA datasets.

A further dimension of interest is the role of *match context*. Playing surface (clay, grass, hard court) and tournament tier (Grand Slam, Mas-

ters, ATP 250) may influence predictability, as certain surfaces favour particular playing styles and higher-tier events tend to have more predictable outcomes due to seeding structures [5]. Whether surface and tournament level provide additional predictive power *beyond* what is already captured by player-strength measures is an empirical question that this project addresses directly.

1.2 Research Question

The core research question of this project is:

How do rating-based (Elo) and statistical learning (logistic regression) models differ in their ability to predict ATP tennis match outcomes, and under what match contexts (such as surface, tournament tier, and ranking-point differences) does one approach show better calibration and predictive accuracy?

1.3 Contributions

This project makes the following contributions:

1. A corrected and reproducible implementation of both logistic regression and Elo models for ATP match prediction, addressing a data leakage issue present in the Part A submission.
2. A systematic comparison of twelve models spanning three approaches—logistic regression, Elo ratings, and Elo-logistic hybrids—evaluated on a common held-out test set using multiple metrics.
3. An empirical investigation of whether surface type, tournament level, and interaction terms contribute significant predictive power beyond player-strength covariates.
4. Evidence that simpler models with better-calibrated probability estimates outperform more complex hybrid specifications, contributing to the broader literature on model complexity in sports prediction.

The complete code for this project is publicly available at:

https://github.com/himanshuade/Project_B_Final_report.git.

2 Work Since Part A

Several substantial improvements and extensions have been made since the Part A progress report.

Data leakage correction. The most critical improvement was a revision to the outcome variable definition. In the original Part A implementation, the player roles were assigned non-randomly based on the match result, which introduced a data leakage issue producing artificially perfect classification. The corrected implementation randomly– assigns player roles using a fixed random seed, ensuring the model faces a genuine prediction task. All Part A models were re-fitted under this corrected scheme.

Eight new hybrid models. Part B extends the project with eight new models that embed Elo rating differences within a logistic regression framework. These include a baseline hybrid, an extended hybrid with surface and tournament-level models, an interaction model allowing Elo effects to vary by surface, combined models using both ranking points and Elo simultaneously, a full extended model with all features and interactions, and three intercept variants testing whether relaxing the zero-intercept constraint improves model calibration..

3 Methods

3.1 Dataset

The dataset comprises all recorded ATP singles matches from 2000 to 2019, sourced from Sackmann’s publicly available ATP match archive [6]. After removing rows with missing values in any of the key fields (player names, rankings, ranking points, surface, tournament level, and score), the final dataset contains 60,154 matches across four surface types (hard, clay, grass, and carpet) and five tournament levels (Grand Slams, Masters 1000, ATP 500, ATP 250, and Tour Finals).

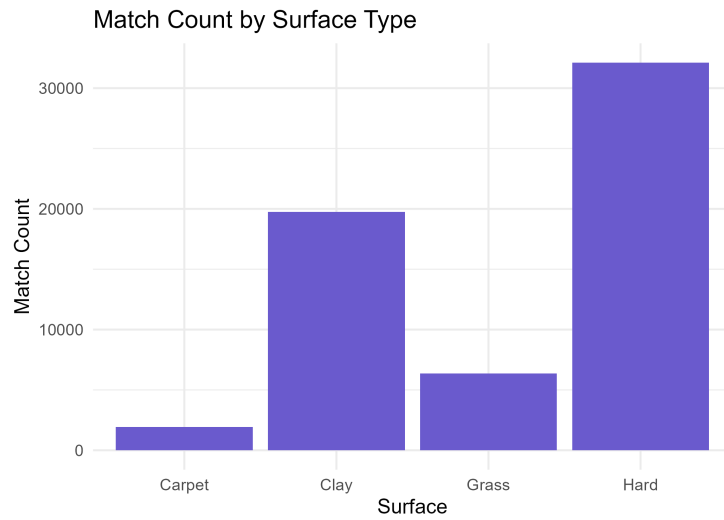


Figure 1: Distribution of matches by surface type

Hard courts account for the majority of matches ($n \approx 33,000$), followed by clay ($n \approx 20,000$), grass ($n \approx 5,000$), and carpet ($n \approx 1,200$). Carpet courts were phased out of the ATP tour during this period, resulting in a small and declining sample.

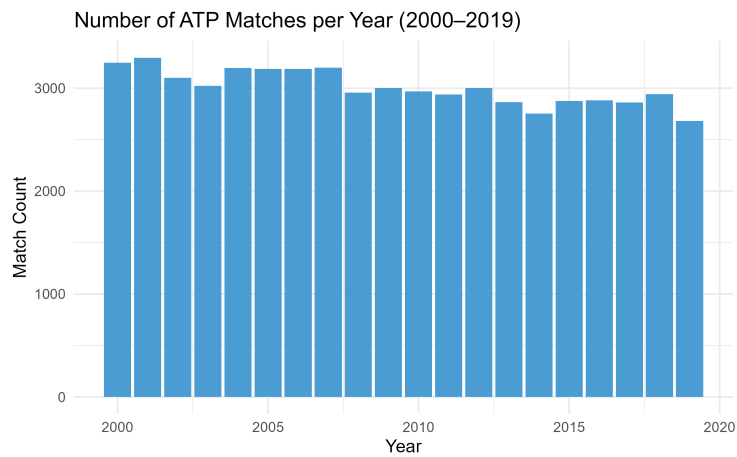


Figure 2: Distribution of ATP matches per year from 2000 to 2019

The distribution of matches per year is approximately uniform at around 3,000 per season, with a slight decline in later years.

3.2 Outcome Variable and Player Assignment

To avoid data leakage, the two players in each match are randomly assigned to the roles of player i and player j using a fixed random seed for reproducibility. Under the random player assignment, the outcome variable is approximately balanced at 50%

The binary outcome variable is defined as

$$y = \begin{cases} 1 & \text{if player } i \text{ won the match,} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Under this scheme, the ranking-point difference $\Delta r = r_i - r_j$ can take positive or negative values with roughly equal frequency, giving the model genuine discriminative signal.

3.3 Train–Test Split

The data are split temporally: matches from 2000–2018 form the training set ($n = 57,474$) and the entire 2019 season forms the test set ($n = 2,680$). This temporal split is consistent with prior literature and avoids data leakage from future matches into training. The 2019 cutoff also avoids COVID-19-related disruptions to the 2020 tour calendar.

3.4 Logistic Regression Models

3.4.1 Baseline Logistic Regression (Model 1)

The baseline logistic regression models the probability that player i wins as a function of the ranking-point difference alone, with no intercept term:

$$\log\left(\frac{P(y = 1)}{1 - P(y = 1)}\right) = \beta_1 \Delta r \quad (2)$$

where $\Delta r = r_i - r_j$ is the difference in ATP ranking points. The zero-intercept constraint ensures that when both players have identical ranking points ($\Delta r = 0$), the predicted probability is exactly 0.5, reflecting an uninformative prior. This specification follows the approach of Klassan and Magnus[3].

3.4.2 Extended Logistic Regression (Model 2)

The extended model adds surface type and tournament level as categorical predictors:

$$\log\left(\frac{P(y=1)}{1-P(y=1)}\right) = \beta_1 \Delta r + \boldsymbol{\gamma}^\top \mathbf{s} + \boldsymbol{\delta}^\top \mathbf{t} \quad (3)$$

where $\mathbf{s}$ is a vector of surface indicator variables and $\mathbf{t}$ is a vector of tournament-level indicators. This model tests whether match context contributes predictive power beyond the ranking-point difference, motivated by evidence that playing conditions influence match outcomes [4].

3.5 Elo Rating Models

The Elo rating system assigns each player a numerical rating that updates after every match. If player i has rating R_i and player j has rating R_j , the expected probability that player i wins is

$$E_i = \frac{1}{1 + 10^{(R_j - R_i)/400}}. \quad (4)$$

After the match, ratings update according to

$$R'_i = R_i + K \cdot (y - E_i) \quad (5)$$

where $y \in \{0, 1\}$ is the outcome and K is the update factor controlling the sensitivity of ratings to individual results.

3.5.1 Baseline Elo (Model 3)

All players begin with a rating of 1500 at the start of the year 2000. Ratings are updated sequentially through each training match with $K = 32$, a standard value used in chess and adopted in prior tennis prediction work [4]. The final ratings at the end of 2018 are then transferred to the test set, where predictions are made using the pre-match Elo-derived probability in Equation (4). Crucially, K is set to zero on the test set so

that ratings are not updated during evaluation.

3.5.2 Surface-Specific K -Factor Elo (Model 4)

Following [5], the K -factor is varied by surface to reflect differing levels of match volatility:

$$K_{\text{surface}} = \begin{cases} 36 & \text{Clay} \\ 32 & \text{Hard} \\ 28 & \text{Carpet} \\ 24 & \text{Grass} \end{cases} \quad (6)$$

The higher value on clay reflects the hypothesis that clay-court results are more sensitive to form differences (longer rallies, fewer service aces), while the lower value on grass reflects greater serve-driven volatility.

3.6 Hybrid Elo–Logistic Models

The hybrid approach uses Elo rating differences as covariates within a logistic regression framework, combining the dynamic responsiveness of Elo with the statistical flexibility of logistic regression. For each match, the Elo difference is defined as $\Delta e = R_i - R_j$, computed from the pre-match ratings obtained during Elo training.

3.6.1 Hybrid Baseline (Model 5)

$$\log\left(\frac{P(y=1)}{1-P(y=1)}\right) = \alpha_1 \Delta e \quad (7)$$

This mirrors the logistic baseline structure but replaces ranking-point difference with Elo difference. Note that Δe enters the logistic regression as a *raw difference*, not as the Elo-derived probability E_i ; feeding in E_i directly would apply the sigmoid transformation twice, since logistic regression already applies a sigmoid internally.

3.6.2 Hybrid Extended (Model 6)

$$\log\left(\frac{P(y=1)}{1-P(y=1)}\right) = \alpha_1 \Delta e + \boldsymbol{\gamma}^\top \mathbf{s} + \boldsymbol{\delta}^\top \mathbf{t} \quad (8)$$

Surface and tournament level are added as contextual covariates alongside the Elo difference.

3.6.3 Interaction Model (Model 7)

$$\log\left(\frac{P(y=1)}{1-P(y=1)}\right) = \alpha_1 \Delta e + \sum_s \lambda_s (\Delta e \times \mathbf{1}_{[\text{surface}=s]}) \quad (9)$$

The interaction term $\Delta e \times \text{surface}$ allows the predictive strength of the Elo difference to vary across surfaces. If, for example, Elo is more informative on clay than on grass, the surface-specific coefficients λ_s will differ.

3.6.4 Combined Models (Models 8–9)

The combined model uses *both* ranking-point difference and Elo difference as predictors:

$$\log\left(\frac{P(y=1)}{1-P(y=1)}\right) = \beta_1 \Delta r + \alpha_1 \Delta e \quad (10)$$

The rationale, following [2], is that ranking points and Elo capture different signals: ranking points reflect official twelve-month standings, while Elo reflects cumulative match-by-match form. Model 9 extends this with surface, tournament level, and the $\Delta e \times \text{surface}$ interaction:

$$\log\left(\frac{P(y=1)}{1-P(y=1)}\right) = \beta_1 \Delta r + \alpha_1 \Delta e + \boldsymbol{\gamma}^\top \mathbf{s} + \boldsymbol{\delta}^\top \mathbf{t} + \sum_s \lambda_s (\Delta e \times \mathbf{1}_{[\text{surface}=s]}) \quad (11)$$

3.6.5 Intercept Models (Models 10–12)

Models 10–12 replicate Models 5, 8, and 9 respectively but include a nonzero intercept term β_0 . All previous models used a zero-intercept constraint to force a 50% prediction when all predictors are zero. Relaxing this constraint allows the model to learn a baseline win probability, which may improve calibration if the true baseline differs from 50%.

3.7 Evaluation Metrics

All models are evaluated on the 2019 test set using four complementary metrics.

Accuracy. Classification accuracy at the 0.5 threshold:

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{[\hat{y}_i=y_i]} \quad (12)$$

Log-loss. The negative mean log-likelihood, which heavily penalises confident incorrect predictions:

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i) \right] \quad (13)$$

where $\hat{p}_i$ is the predicted probability of a win for player i . Predicted probabilities are clipped to $[\epsilon, 1-\epsilon]$ with $\epsilon = 10^{-15}$ for numerical stability. A no-skill baseline (predicting 0.5 for every match) yields a log-loss of $\log 2 \approx 0.693$.

Brier score. The mean squared error of the probabilistic predictions [4]:

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2 \quad (14)$$

Calibration curves. Predicted probabilities are binned into ten equal-width intervals $[0, 0.1], (0.1, 0.2], \dots, (0.9, 1.0]$. Within each bin, the mean predicted probability is plotted against the empirical win rate. A perfectly calibrated model would produce points lying on the 45° diagonal. Some bins may contain few observations, particularly at the extremes, which can produce noisy calibration estimates.

ROC–AUC. The area under the receiver operating characteristic curve measures discriminative ability independently of the classification threshold.

4 Results

4.1 Part A Models

4.1.1 Logistic Regression

The baseline logistic model (Model 1) estimates a coefficient of $\hat{\beta}_1 = 5.138 \times 10^{-4}$ for the ranking-point difference ($z = 64.66$, $p < 0.001$). This coefficient is highly significant and indicates that each additional ranking point in favour of player i increases the log-odds of winning by approximately 5.1×10^{-4} . On the test set, Model 1 achieves an accuracy of 61.4%, log-loss of 0.655, Brier score of 0.229, and AUC of 0.669.

The extended logistic model (Model 2) adds surface and tournament level as predictors. The coefficient on ranking-point difference stays almost the same ($\hat{\beta}_1 = 5.138 \times 10^{-4}$). An ANOVA test shows that adding surface ($p = 0.18$) and tournament level ($p = 0.14$) does not significantly improve the model once ranking-point difference is already included. Test-set performance is nearly identical to the baseline: accuracy 61.2%, log-loss 0.656, Brier score 0.229, and AUC 0.669.

4.1.2 Elo Ratings

The baseline Elo model (Model 3) with $K = 32$ achieves accuracy of 60.9%, log-loss of 0.670, Brier score of 0.237, and a ten-bin calibration curve that tracks the diagonal reasonably well, with the main deviations occurring in the lowest-probability bin (predicted 6.9%, observed 12.5%) and the 70–80% bin (predicted 75.1%, observed 59.8%).

The surface-specific K -factor model (Model 4) shows a marginal improvement: accuracy increases to 61.5% and log-loss decreases very slightly to 0.670. The Brier score is essentially unchanged at 0.237. The calibration curve shows a similar pattern to the baseline Elo but with slightly better tracking in the upper probability bins.

4.2 Part B: Hybrid Models

4.2.1 Hybrid Baseline and Extended (Models 5–6)

The hybrid baseline (Model 5) estimates a coefficient of $\hat{\alpha}_1 = 8.323 \times 10^{-3}$ for the Elo difference ($p < 0.001$). This is much larger than the ranking-point coefficient because the two predictors are on different scales – Elo differences are typically in the hundreds, while ranking-point differences can reach tens of thousands. On the test set, accuracy is 60.9% and AUC is 0.665, which is competitive with the Part A models. However, log-loss is noticeably worse at 0.730 and Brier score is 0.252, meaning the predicted probabilities are less reliable.

The extended hybrid (Model 6) adds surface and tournament level but does not improve performance. An ANOVA test shows that surface ($p = 0.57$) and tournament level ($p = 0.08$) are not significant once the Elo difference is already in the model. Test-set results are nearly identical: accuracy 61.0%, log-loss 0.731, Brier 0.252.

4.2.2 Interaction Model (Model 7)

Model 7 includes $\Delta e \times$ surface interaction terms. The surface-specific coefficients for clay ($+0.00025$, $p = 0.15$), grass (-0.00036 , $p = 0.15$), and carpet ($+0.00062$, $p = 0.20$) are all statistically non-significant, with the hard-court coefficient dropped due to collinearity. Test-set performance is effectively unchanged: accuracy 60.9%, log-loss 0.730, Brier 0.252.

4.2.3 Combined Models (Models 8–9)

Model 8 includes both Δr and Δe as predictors. Notably, the ranking-point coefficient reverses sign ($\hat{\beta}_1 = -2.868 \times 10^{-4}$, $p < 0.001$) while the Elo coefficient increases in magnitude ($\hat{\alpha}_1 = 1.042 \times 10^{-2}$, $p < 0.001$). The negative coefficient on ranking points suggests that, when Elo difference is already in the model, higher ranking points are associated with a *lower* probability of winning. This counterintuitive result is a sign of collinearity: ranking points and Elo ratings both measure player strength, and when both are included together, the individual coefficients become unstable.

Despite fitting the training data well (lowest AIC of 59,277), this model performs worst on the test set: log-loss rises to 0.764 and Brier score to 0.261, with AUC dropping to 0.654. This suggests that combining two highly correlated predictors leads to overfitting.

The full extended model (Model 9) adds surface, tournament level, and interaction terms on top of Model 8. An ANOVA test shows that surface ($p = 0.69$), tournament level ($p = 0.08$), and the interaction ($p = 0.20$) remain non-significant. Test-set performance is the same as Model 8: accuracy 60.5%, log-loss 0.764, Brier 0.261, AUC 0.653.

4.2.4 Intercept Models (Models 10–12)

Models 10–12 relax the zero-intercept constraint on Models 5, 8, and 9. In Model 10, the estimated intercept is $\hat{\beta}_0 = -0.0108$ ($z = -1.09$, $p = 0.28$), which is not statistically significant. This confirms that the true baseline win probability is indeed close to 50%, as expected from the random player assignment. Similarly, the intercept in Model 11 is -0.0093 ($p = 0.35$) and in Model 12 is 0.0128 ($p = 0.81$), both non-significant. Removing the zero-intercept constraint does not improve calibration: all three intercept models produce test-set log-loss, Brier, and accuracy values virtually identical to their constrained counterparts (Table 1).

4.3 Model Comparison

Table 1 presents the full comparison of all twelve models on the 2019 test set.

Table 1: Predictive performance of all twelve models on the 2019 ATP test set ($n = 2,680$). Lower log-loss and Brier score indicate better probabilistic calibration. The best value in each column is shown in bold.

#	Model	Accuracy	Log-Loss	Brier	AUC
<i>Part A - Logistic Regression</i>					
1	Baseline (Δr only)	0.614	0.655	0.229	0.669
2	Extended (+ s , t)	0.612	0.656	0.229	0.669
<i>Part A — Elo Ratings</i>					
3	Baseline ($K = 32$)	0.609	0.670	0.237	—
4	Surface- K	0.615	0.670	0.237	—
<i>Part B — Hybrid (no intercept)</i>					
5	Δe only	0.609	0.730	0.252	0.665
6	$\Delta e + \mathbf{s} + \mathbf{t}$	0.610	0.731	0.252	—
7	$\Delta e + \Delta e \times \mathbf{s}$	0.609	0.730	0.252	—
8	$\Delta r + \Delta e$	0.604	0.764	0.261	0.654
9	Full ($\Delta r + \Delta e + \mathbf{s} + \mathbf{t} + \Delta e \times \mathbf{s}$)	0.605	0.764	0.261	0.653
<i>Part B — Hybrid (with intercept)</i>					
10	$\beta_0 + \Delta e$	0.609	0.731	0.252	—
11	$\beta_0 + \Delta r + \Delta e$	0.603	0.764	0.261	—
12	$\beta_0 +$ Full	0.605	0.764	0.261	—

4.4 Visualisations

Three key figures summarise the comparative results.

Figure 3 presents the overlaid ROC curves for the three core modelling approaches: the logistic baseline (Model 1), the Elo baseline (Model 3), and the hybrid baseline (Model 5). All three curves are nearly indistinguishable, with AUC values ranging from 0.665 to 0.669, indicating that discriminative ability is similar across paradigms. The curves sit well above the diagonal, confirming that all models provide meaningful predictive signal.

Figure 4 displays a bar chart of log-loss across all twelve models, with a red dashed line at $\log 2 \approx 0.693$ indicating the no-skill baseline. The

Part A logistic models (blue bars) sit comfortably below this line, while several of the Part B hybrid and combined models (purple and orange bars) approach or exceed it, indicating calibration close to or worse than random guessing.

Figure 5 shows side-by-side calibration curves for the logistic baseline and the hybrid baseline. The logistic baseline exhibits points that track the 45° diagonal across the full range of predicted probabilities, with relatively even spacing. The hybrid baseline, by contrast, shows more pronounced deviations from the diagonal, particularly in the lower probability bins where predictions are too extreme relative to observed outcomes.

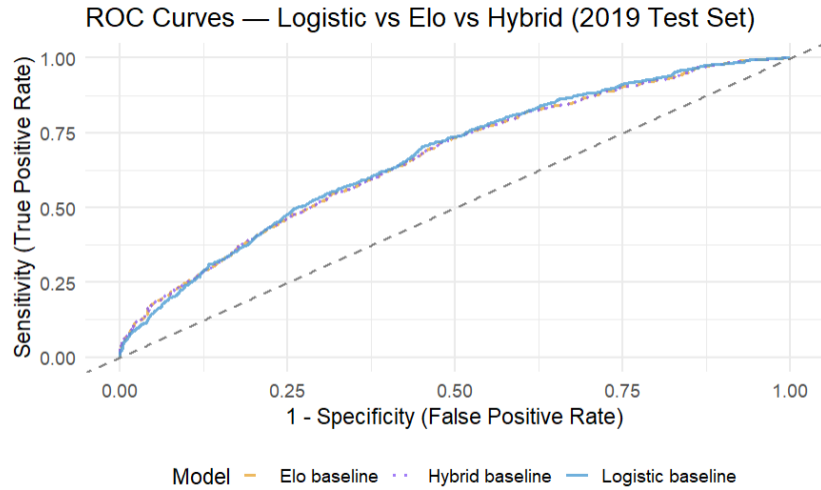


Figure 3: ROC curves for the logistic baseline (AUC = 0.669), Elo baseline, and hybrid baseline (AUC = 0.665), evaluated on the 2019 test set. All three curves are nearly overlapping, indicating comparable discriminative ability across paradigms.

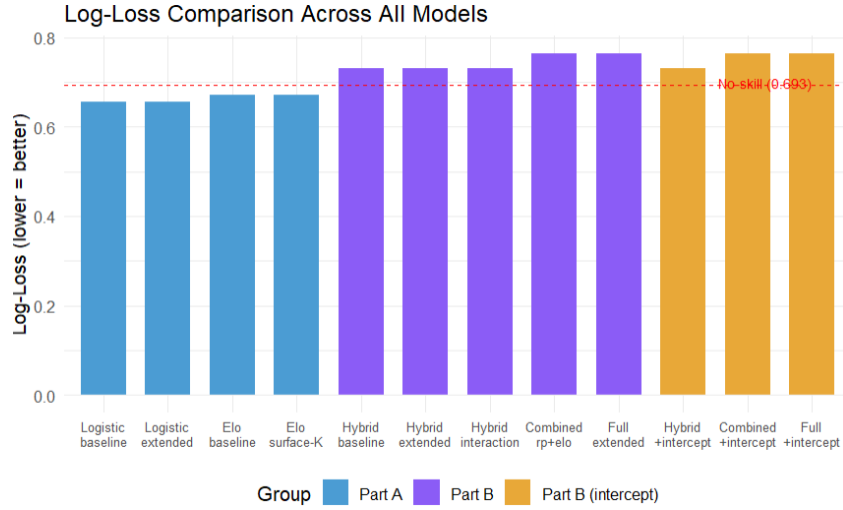


Figure 4: Log-loss comparison across all twelve models. Blue bars represent Part A models, purple bars represent Part B hybrid models without intercept, and orange bars represent Part B models with intercept. The red dashed line at 0.693 marks the no-skill baseline. Lower values indicate better probabilistic calibration.

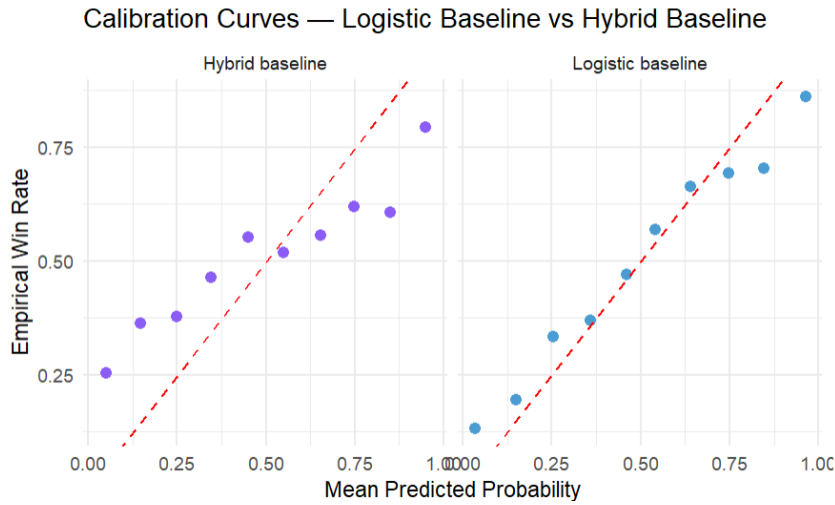


Figure 5: Calibration curves comparing the logistic baseline (left) and hybrid baseline (right). Each point represents one of ten equal-width probability bins; the x -axis is the mean predicted probability within each bin and the y -axis is the empirical win rate. Points on the red dashed diagonal indicate perfect calibration. The logistic baseline tracks the diagonal more closely, while the hybrid model shows systematic deviation.

5 Discussion

5.1 Ranking Points versus Elo Ratings

The results reveal a clear pattern: when used as a standalone predictor within a logistic regression, the ranking-point difference produces better-calibrated probability estimates than the Elo difference. The logistic baseline (Model 1) achieves a log-loss of 0.655, while the hybrid baseline (Model 5) achieves 0.730 - a substantial degradation despite achieving comparable accuracy (61.4% versus 60.9%).

This result might seem surprising, since Elo ratings are designed to give good win probabilities on their own (Equation (4)). The problem is that Model 5 does not use the Elo probability directly – instead, it feeds the raw Elo *difference* into a logistic regression. The logistic regression then applies its own transformation on top, which ends up pushing the predicted probabilities too far toward 0 and 1. Ranking-point differences do not have this issue because they are raw numbers that the logistic regression can map to probabilities cleanly.

The calibration curves in Figure 5 show this clearly: the hybrid model predicts around 0.2 for some matches where the actual win rate is closer to 0.35, and predicts around 0.8 where the actual win rate is closer to 0.65. The predictions are too extreme in both directions.

5.2 The Role of Surface and Tournament Level

A consistent finding across all model specifications is that surface type and tournament level do not contribute statistically significant predictive power once a player-strength predictor (either Δr or Δe) is in the model. In the logistic extended (Model 2), surface has $p = 0.18$ and tournament level has $p = 0.14$. In the hybrid extended (Model 6), the corresponding values are $p = 0.57$ and $p = 0.08$. The $\Delta e \times \text{surface}$ interaction in Model 7 is similarly non-significant.

This result does not mean that surface and tournament level are irrelevant to match outcomes - they clearly are, as playing styles and seeding structures differ across these contexts. Rather, it suggests that these

effects are already *implicitly encoded* in the player-strength measures. Players' ranking points and Elo ratings are shaped by their surface preferences and the level of tournaments they enter: a clay-court specialist will accumulate more points from clay events, and their Elo will reflect their performance on their preferred surface. Adding these variables explicitly therefore provides limited additional information for prediction.

5.3 Collinearity in Combined Models

The combined models (Models 8–9 and 11–12) produce the worst calibration in the study, with log-loss values of 0.764. The sign reversal on the ranking-point coefficient in the combined specification – negative when Δe is also present – is a hallmark of multicollinearity. Ranking points and Elo ratings are both measures of the same latent construct (player strength), and their high correlation (Δr and Δe are derived from overlapping match histories) means the model cannot reliably separate their individual contributions.

Despite achieving the lowest in-sample AIC (59,277 for Model 8 versus 72,568 for Model 1), the combined models exhibit classic overfitting: improved fit on training data but degraded generalisation. This is a practically important result for tennis prediction: combining multiple correlated strength measures does not improve out-of-sample performance and, in this case, actively harms it.

5.4 Intercept Constraint

Models 10–12 test whether allowing a nonzero intercept improves calibration. The intercept is not statistically significant in any of the three specifications, and removing the zero-intercept constraint does not improve test-set performance. This confirms that the random player assignment produces a balanced dataset where the baseline win probability is close to 50%, making the zero-intercept constraint appropriate.

5.5 Relation to Prior Literature

The overall accuracy range of 60–62% across all models is consistent with published benchmarks for ATP match prediction. Kovalchik[4] reported similar accuracy levels across a variety of methods and concluded that tennis outcomes are inherently difficult to predict due to the influence of unobservable factors such as motivation, fatigue, and in-match tactical adjustments.

The finding that Elo-based models do not substantially outperform ranking-point models aligns with the broader conclusion in Kovalchik[4] that no single method dominates across all evaluation criteria. The calibration advantage of the ranking-point logistic model over the Elo-logistic hybrid adds nuance to this finding: while discriminative ability is similar (AUC within 0.004), the quality of the predicted *probabilities* differs meaningfully.

The non-significance of surface and tournament level is consistent with Klaassen and Magnus [3], who found that ranking differences account for the majority of predictive power in tennis. Angelini et al.[1] similarly found that their Weighted Elo extension, while theoretically motivated, produced only marginal gains over a standard Elo baseline.

5.6 Limitations and Future Work

This study could be extended in several directions. The Elo K -factors used in this project were drawn from prior literature; a natural next step would be to optimise these values through cross-validation to find the best fit for ATP data specifically. The models could also be enriched with additional features available in the dataset, such as player age, head-to-head records, or recent form, which may capture aspects of player performance not reflected in aggregate strength measures. Evaluating across multiple test seasons rather than a single year (2019) would provide a more robust assessment of model performance across different competitive conditions. Finally, maintaining surface-specific Elo ratings separate ratings for clay, grass, and hard courts could more directly capture how players perform differently across surfaces, and is a –promising

direction for improving the hybrid modelling approach.

6 Conclusion

This project compared twelve models across three different approaches—logistic regression, Elo ratings, and Elo–logistic hybrids, for predicting ATP tennis match outcomes. Using 60,154 matches from 2000–2019 and a temporally held-out 2019 test set, we find that all models achieve accuracy in the range of 60–62%, consistent with established benchmarks in the literature. The key differentiator between models is *calibration quality* (log-loss and Brier score) rather than raw accuracy or discriminative ability (AUC).

The simplest model: logistic regression with ranking-point difference as the sole predictor, achieves the best calibration (log-loss 0.655, Brier 0.229), outperforming all hybrid and combined specifications. Surface type, tournament level, and interaction terms do not contribute statistically significant predictive power once a player-strength predictor is included, suggesting these contextual factors are already captured within ranking points and Elo ratings. Combining ranking points and Elo differences in a single model produces worse calibration on the test set due to high predictor correlation, despite fitting the training data well.

Overall, these results suggest that for ATP match prediction, a simpler model with well-calibrated probabilities can outperform more complex specifications. The logistic regression using ranking-point difference alone proved to be the most reliable approach, and adding more predictors or combining different strength measures did not improve performance on unseen data.

Future work could extend this analysis by optimising K -factors via cross-validation, maintaining surface-specific Elo ratings, incorporating additional player-level features such as age and head-to-head records, and evaluating across multiple test seasons for more robust estimates.

Acknowledgements

I would like to thank *Anugus Lewis* and *Mikaela Fudolig* for their supervision and feedback across both parts of this project.

References

- [1] G. Angelini, V. Candila, and L. De Angelis. Weighted Elo rating for tennis match predictions. *European Journal of Operational Research*, 297(1):120–132, 2022.
 - [2] L. M. Hvattum and H. Arntzen. Using Elo ratings for match result prediction in association football. *International Journal of Forecasting*, 26(3):460–470, 2010.
 - [3] F. J. G. M. Klaassen and J. R. Magnus. Forecasting the winner of a tennis match. *European Journal of Operational Research*, 148(2):257–267, 2003.
 - [4] S. A. Kovalchik. Searching for the GOAT of tennis win prediction. *Journal of Quantitative Analysis in Sports*, 12(3):127–138, 2016.
 - [5] S. A. Kovalchik. Extension of the Elo rating system to margin of victory. *International Journal of Forecasting*, 36(4):1329–1341, 2020.
 - [6] J. Sackmann. Tennis ATP match data [dataset], 2023.
-